PredictX-HARBOR: Reliability-Conditioned Biomarker Graphs, Heterogeneous Ensembles, and Value-of-Information Test Acquisition for Multi-Disease Prediction

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Abstract: **Clinical biomarker data differ from generic tables: units, specimens, assays and collection times change what a value means, whether a test was ordered carries information, and each disease is observed in its own cohort. We present PredictX-HARBOR, a multi-disease prediction system built on a typed, availability-aware biomarker representation. Its learning core, HARBOR, is a graph encoder that separates a value-message graph over observed biomarkers from a missingness-state graph over all biomarkers and mixes them with an availability-dependent gate; graph sparsity is regularized by attention entropy, and we prove that the ℓ_1 penalty commonly used for this purpose has identically zero gradient under softmax attention. Predictions are produced by an equal-weight ensemble of TabPFN v2, CatBoost, penalized logistic regression and HARBOR, which provably scores at least as well as its average member on proper scoring rules and supports cross-site training on cohorts with different test availability, and a value-of-information controller selects the next test. On five disease cohorts (coronary disease, diabetes, breast malignancy, liver disease and chronic kidney disease), evaluated with 5×5 repeated cross-validation and corrected intervals against seven literature-standard methods and two state-of-the-art learners, PredictX attains the highest AUROC for coronary disease (0.915), is within 0.002 of the best learner on the other four cohorts, is significantly better in 12 and significantly worse in none of 50 paired comparisons, and has a lower Brier score than the best literature-standard method on all five cohorts. Under the protocol of the published TabPFN benchmark it exceeds the best published breast (0.9953 versus 0.9940) and liver (0.7438 versus 0.7412) results. For coronary disease the acquisition controller retains 98% of full-panel AUROC at 53% of the testing cost. The deployed models score the five clinical input forms with AUROC 0.741–0.996 and are served through an HTTP interface released with the full experimental code.**

Index Terms: **clinical prediction, biomarker graphs, missing data, tabular foundation models, ensembles, active feature acquisition**

I. Introduction

Routine clinical laboratory and examination data carry most of the information used to recognize chronic disease, yet they differ from the tables on which general-purpose learning methods are developed. A serum value acquires its meaning from its unit, specimen, assay and collection time; whether a test was ordered at all reflects the clinician's suspicion and is therefore informative; and each disease is studied in its own cohort, with its own feature set and label definition. A system that serves several diseases must represent this heterogeneity explicitly, predict each endpoint as accurately as the available biomarkers allow, and help decide which additional test is worth its cost.

Deep tabular models have only recently become competitive with gradient-boosted trees on such data [5, 2], and the strongest small-data learner to date is a prior-fitted transformer that performs Bayesian inference in context [14, 15]. Clinical graph models inject structural priors into attention [8], and missing-data theory shows that the Bayes-optimal predictor depends on the missingness pattern whenever values are missing not at random [10, 11]. These ingredients have so far been studied separately.

We present PredictX-HARBOR, a multi-disease prediction system built on a typed, availability-aware biomarker representation shared by three components: a graph encoder, a heterogeneous ensemble and a test-acquisition controller. The contributions are as follows.

1. **HARBOR**, a reliability-conditioned biomarker graph encoder in which a value-message graph over observed biomarkers and a missingness-state graph over all biomarkers are mixed by an availability-dependent gate (Section IV). Unobserved biomarkers retain their concept identity, graph sparsity is regularized by attention entropy, and we prove that the ℓ_1 attention penalty has identically zero gradient under softmax normalization (Proposition 2).
2. **The PredictX ensemble**, an equal-weight combination of TabPFN, CatBoost, penalized logistic regression and HARBOR with a proper-score guarantee (Proposition 4), together with cross-site training on cohorts that record the same concepts with different test availability (Section V).
3. **A value-of-information test-acquisition controller** with a non-parametric conditional surrogate and a pattern-complete predictor, applied to sequential ordering of exercise electrocardiography, fluoroscopy and thallium scintigraphy (Section VI).
4. **An evaluation on five disease cohorts** against seven literature-standard methods and two state-of-the-art learners with corrected repeated cross-validation, a reproduction of the strongest published benchmark protocol, and abla-

tions of every component (Sections VIII–IX).

5. **A deployed web service** that maps the five clinical input forms to the trained ensembles and returns a risk score for each endpoint, released with the full experimental code (Sections VII and XI).

II. Related Work

Deep learning for tabular clinical data. Feature-token transformers (TabTransformer, FT-Transformer [1, 2]), attentive feature selection (TabNet [3]) and graph attention [4] learn interactions among heterogeneous columns, while gradient-boosted trees remain strong reference learners on small tables [5, 6, 7]. EHR-specific models add structure to attention: the Graph Convolutional Transformer uses encounter structure as an attention prior [8], and Graphormer adds structural biases to attention logits [9]. HARBOR combines a clinical prior on the attention logits with an explicit availability channel.

Learning with missing values. When values are missing not at random (MNAR), the Bayes predictor depends on the missingness pattern [10]. NeuMiss derives network architectures from the conditional expectation under missingness [11], Le Morvan et al. characterize when impute-then-predict is consistent [12], and GRU-D models informative missingness in time series through learned decay [13]. HARBOR propagates availability through a dedicated message-passing channel so that the mask modulates value propagation.

In-context tabular foundation models and ensembles. TabPFN is a transformer meta-trained on synthetic data drawn from a prior over structural causal models; it approximates the posterior predictive distribution in a single forward pass [14], and its second version extends the approach to larger tables and categorical features [15]. Combining dissimilar learners reduces variance and guards against misspecification [16, 17], and proper scoring rules make such combinations analyzable [18].

Active feature acquisition. Budgeted, sequential test ordering has been formulated as reinforcement learning with set encoders [19] and as information-theoretic acquisition with partial variational autoencoders [20]. Our controller uses a nonparametric conditional surrogate and a decision-theoretic utility, which suits cohorts of a few hundred patients.

Evaluation practice. Reported accuracies on the public disease cohorts vary widely, in part because of leakage through preprocessing or oversampling before splitting and through pooling of overlapping cohorts [21]. We re-implement the standard comparators under a single leakage-free protocol, follow TRIPOD+AI reporting guidance [22], and reproduce the strongest published benchmark protocol where one exists.

III. Problem Formulation

Typed biomarkers. Each measurement is a typed record $b_j = (c_j, v_j, u_j, t_j, s_j, a_j, q_j, m_j, \ell_j)$ with clinical concept c, value v, unit u, collection time t, specimen or modality s, assay a, quality metadata q, availability state m, and provenance ℓ. Unit conversion is a versioned map $\tilde{v}_{ij} = T_{c_j, u_j \to u_j^*}(v_{ij}; \theta_T)$ that is defined for dimensionally and clinically compatible specimen–unit pairs; other inputs are rejected at ingestion. Availability is a first-class variable: $m_{ij} \in \{1, 0\}$ with a typed reason μ_{ij} (never measured, invalid assay, withheld, out of window, inapplicable) whenever the source records one.

Endpoints. Status classification estimates $P(Y_d = 1 \mid X_{\le t})$ for a cohort-defined label; incident risk estimates $P(T_d \le t + \tau \mid T_d > t, \mathcal{H}_t)$ and requires a disease-free baseline cohort. The runtime stores the endpoint type with every prediction, and records without a verified label for d are excluded from the loss for d.

Learning problem. For endpoint d with labeled set $\mathcal{I}_d$, we observe $(\tilde{\boldsymbol{v}}_i \odot \boldsymbol{m}_i, \boldsymbol{m}_i, y_{id})$ and seek a risk score $p_{id} = P(Y_d = 1 \mid \tilde{\boldsymbol{v}}_i \odot \boldsymbol{m}_i, \boldsymbol{m}_i)$ that uses availability as information, is defined for every availability pattern that can occur at decision time, and supports a sequential decision about which unobserved biomarker to acquire next.

IV. HARBOR: Reliability-Conditioned Biomarker Graph Encoder

A. Typed tokenization and reliability gate

For patient i and biomarker j an observed token combines concept identity with value, age, specimen and quality channels,

$$\begin{aligned} h_{ij}^{(0)} = E_{c_j} + f_{\mathrm{val}}(\tilde{v}_{ij}) + f_{\mathrm{time}}(\Delta t_{ij}) \\ + f_{\mathrm{type}}(s_j) + f_{\mathrm{qual}}(q_{ij}, m_{ij}), \end{aligned} \tag{1}$$

where f_{val} is an affine embedding of the quantile-normalized value for numeric concepts and a learned table for categorical ones. A bounded reliability gate scales the evidence a token may emit, and an absent biomarker becomes a missingness-state token that keeps its concept identity:

$$\begin{aligned} r_{ij} &= \sigma\Big(w_r^\top \big[h_{ij}^{(0)}; q_{ij}; m_{ij}; \Delta t_{ij}\big] + b_r\Big), \\ \bar{h}_{ij} &= m_{ij}\, h_{ij}^{(0)} + (1 - m_{ij})\big(E_{c_j} + e_{\mu_{ij}}\big). \end{aligned} \tag{2}$$

Keeping E_{c_j} in the missing token preserves concept identity, so an unmeasured creatinine and an unmeasured age are represented by different vectors. The bias b_r is initialized at 2 so that gates start open ($\sigma(2) \approx 0.88$).

B. Two-channel message passing

Let $\bar{H}_i \in \mathbb{R}^{J \times d}$ stack the tokens. HARBOR maintains two graphs per layer. The *value graph* attends only among observed biomarkers and transmits gated value evidence; the *missingness-state graph* attends among all eligible biomarkers using availability embeddings that contain no physiological value:

$$H^v = \boldsymbol{m} \odot \boldsymbol{r} \odot H, \tag{3}$$

$$H^m = E_{\boldsymbol{c}} + \boldsymbol{m} \odot e_{\mathrm{obs}} + (1 - \boldsymbol{m}) \odot e_{\boldsymbol{\mu}}, \tag{4}$$

$$\begin{aligned} A^v = \mathrm{softmax}\Big(\tfrac{(H^v Q_v)(H^v K_v)^\top}{\sqrt{d}} + \alpha_d A_d^{\mathrm{prior}} \\ + B^v\Big), \end{aligned} \tag{5}$$

$$\begin{aligned} A^m = \mathrm{softmax}\Big(\tfrac{(H^m Q_m)(H^m K_m)^\top}{\sqrt{d}} \\ + B^m\Big), \end{aligned} \tag{6}$$

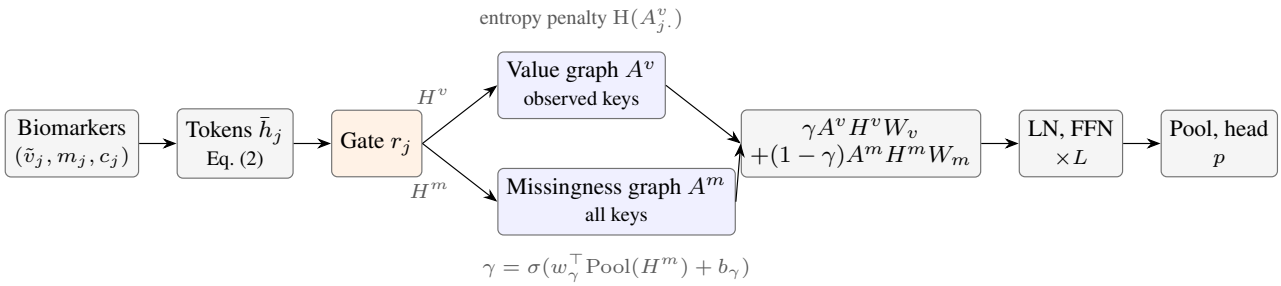


Fig. 1. One HARBOR layer. Gated value tokens H^v propagate over the value graph among observed biomarkers; availability tokens H^m propagate over the missingness-state graph among all biomarkers; an availability-dependent gate γ mixes the two messages.

where $B^v_{jk} = -\infty$ if biomarker k is unobserved or ineligible and B^m masks only ineligible biomarkers. The layer update mixes the two channels with an availability-dependent gate:

$$\begin{aligned} \gamma &= \sigma\big(w_\gamma^\top \mathrm{Pool}(H^m) + b_\gamma\big), \\ H' &= \mathrm{LN}\big(H + \gamma A^v H^v W_v + (1-\gamma) A^m H^m W_m\big), \\ H^+ &= \mathrm{LN}\big(H' + \mathrm{FFN}(H')\big). \end{aligned} \tag{7}$$

Because γ is a function of the availability pattern, the effective interaction topology changes with what has been measured: a patient with a complete laboratory panel is scored mainly through value interactions, and a sparsely tested patient through the pattern of testing, which carries information whenever ordering decisions depend on clinical suspicion. Disease experts pool the final layer and apply a two-layer head, $p_{id} = \sigma\big(g_d(\mathrm{Pool}_{j\in\mathcal{J}_a} h^{(L)}_{ij}; \theta_d)\big)$, trained only on records with a verified label for d.

Proposition 1 (Cost)**.** *One HARBOR layer costs $O\big(J^2 d + J d^2\big)$ per patient for J biomarkers and width d, the same order as a single-head feature-token transformer; the second channel doubles the constant, not the order.*

C. Training objective and graph regularization

The objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{task}} + \lambda_g \mathcal{L}_{\mathrm{graph}} + \lambda_r \mathcal{L}_{\mathrm{rel}} + \lambda_c \mathcal{L}_{\mathrm{consistency}}, \tag{8}$$

with $\mathcal{L}_{\mathrm{task}} = \sum_d \sum_{i\in\mathcal{I}_d} w_{id}\, \ell(y_{id}, p_{id})$ the label-masked binary cross-entropy, $\mathcal{L}_{\mathrm{rel}} = \sum_{(i,j)\in\mathcal{Q}} (r_{ij} - q^*_{ij})^2$ on records with independent quality targets, and $\mathcal{L}_{\mathrm{consistency}} = \sum_k \mathbb{E}[\max(0, g_k(\hat{y}, X))^2]$ for a reviewed rule catalog (unit ranges, temporal ordering, admissible monotonicities). The graph term is

$$\mathcal{L}_{\mathrm{graph}} = \sum_{d,i} \left[\frac{1}{|\mathcal{V}_i|} \sum_{j\in\mathcal{V}_i} \mathrm{H}\big(A^v_{i,j\cdot}\big) + \kappa \big\| A^v_i - \bar{A}^{\mathrm{prior}}_d \big\|_F^2 \right], \tag{9}$$

where $\mathrm{H}(a) = -\sum_k a_k \log a_k$, $\mathcal{V}_i$ is the observed set and $\bar{A}^{\mathrm{prior}}_d$ the row-normalized reviewed prior. The choice of entropy rather than an ℓ_1 norm follows from two elementary facts.

Proposition 2 (ℓ_1 degeneracy under softmax)**.** *If every row of $A \in \mathbb{R}^{n\times n}$ is a softmax of logits Z, then $\|A\|_1 = n$ for all Z and $\nabla_Z \|A\|_1 = 0$. An ℓ_1 sparsity penalty on attention therefore has no effect on training.*

Proof. Entries of a softmax row are positive and sum to one, so $\|A\|_1 = \sum_j \sum_k A_{jk} = \sum_j 1 = n$, a constant in Z. □

Proposition 3 (Entropy as a sparsity measure)**.** *For a distribution a over n keys, $0 \le \mathrm{H}(a) \le \log n$, with $\mathrm{H}(a) = 0$ iff a is one-hot and $\mathrm{H}(a) = \log n$ iff a is uniform. With $a = \mathrm{softmax}(z)$, $\partial \mathrm{H}/\partial z_k = -a_k(\log a_k + \mathrm{H}(a))$, which vanishes for every k only when a is uniform on its support (one-hot rows included).*

Proof. The bounds are standard properties of Shannon entropy. For the gradient, $\partial a_j/\partial z_k = a_j(\delta_{jk} - a_k)$ and $\partial \mathrm{H}/\partial a_j = -(\log a_j + 1)$; combining and using $\sum_j a_j = 1$ gives the stated expression. □

Numerically, on random 6×6 logits the largest ℓ_1 gradient entry is 6×10^{-8} (floating-point noise), against 7×10^{-2} for the entropy term, and 300 entropy-descent steps raise the largest weight per row from 0.24–0.65 to 0.99. Sparsemax and α-entmax [23, 24] are alternatives that produce exact zeros.

V. PredictX: Heterogeneous Ensemble and Cross-Site Training

A. Ensemble members

The deployed predictor for each endpoint averages four learners whose inductive biases differ sharply:

$$p^{\mathrm{ENS}}_i = \tfrac{1}{4}\big(p^{\mathrm{PFN}}_i + p^{\mathrm{CAT}}_i + p^{\mathrm{LR}}_i + p^{\mathrm{HARBOR}}_i\big). \tag{10}$$

TabPFN (version 2, eight estimators) [14, 15] approximates the Bayesian posterior predictive under a prior over structural causal models, conditioning on the whole training set in context. *CatBoost* [7] is gradient boosting with ordered target statistics, which limits prediction shift on small samples. *Logistic regression* with median imputation, missing-value indicators and an ℓ_2 penalty chosen by inner cross-validation contributes a low-variance additive model. *HARBOR* contributes availability-conditioned interactions; it is itself a five-member deep ensemble [17], each member early-stopped on a different stratified 20% split. Weights are fixed at $1/4$, which keeps the number of fitted ensemble parameters at zero on cohorts of a few hundred patients.

Proposition 4 (Proper-score guarantee)**.** *Let S be the Brier score or the log loss. For any member forecasts $p_1, \ldots, p_K \in$*

$(0,1)$ *and outcome* $y \in \{0,1\}$, $S\big(\frac{1}{K}\sum_k p_k, y\big) \le \frac{1}{K}\sum_k S(p_k, y)$*, with equality iff all* p_k *are equal. The gap for the Brier score equals the variance of the member forecasts,* $\frac{1}{K}\sum_k (p_k - \bar{p})^2$.

Proof. Both scores are convex in p, so the inequality is Jensen's. For the Brier score, $\frac{1}{K}\sum_k (p_k - y)^2 = (\bar{p} - y)^2 + \frac{1}{K}\sum_k (p_k - \bar{p})^2$. □

The ensemble therefore scores at least as well as its average member on both proper scores, and the improvement grows with member disagreement, which heterogeneous inductive biases supply.

B. Cross-site auxiliary training

When additional cohorts record the same concepts with different availability patterns, HARBOR's typed tokens allow them to be pooled without imputation. For the coronary endpoint we train on the target cohort plus the Hungarian, Swiss and VA Long Beach cohorts [25], which share the same concept schema but are missing up to 98% of some tests. Evaluation uses held-out patients of the target cohort.

VI. Value-of-Information Test Acquisition

A. Controller

The state of a patient is the observed set $\mathcal{O}_t$ and the current prediction p_t. Tests are blocks of biomarkers $a \in \mathcal{A}_t$ with cost $C(a)$; the controller orders

$$a_t^* = \arg\max_{a \in \mathcal{A}_t} \Big[\mathbb{E}\{U(p_{t+1}) \mid a, \mathcal{O}_t\} - U(p_t) - \lambda C(a) \Big] \quad (11)$$

if the bracket is positive and stops otherwise. We study two utilities: the decision utility $U_{01}(p) = -\min(p, 1-p)$, the negative Bayes risk of the best binary decision, and the information utility $U_{\mathrm{H}}(p) = p \log p + (1-p)\log(1-p)$, the negative predictive entropy, for which Eq. (11) is expected information gain minus cost.

Conditional surrogate. The expectation requires $p(z_a \mid \mathcal{O}_t)$ for the unseen block z_a. We use the empirical distribution of z_a among the $K = 25$ training patients nearest to the current patient on the standardized observed biomarkers. Each candidate test is then scored by K forward passes of the predictor for availability pattern $\mathcal{O}_t \cup a$.

Pattern-complete predictor. The policy queries predictions for every intermediate availability pattern. We train one predictor per pattern ($2^{|\mathcal{A}|}$ models; eight for three tests), each on exactly the biomarkers it will see.

VII. System Architecture and Deployment

A. Architecture

Fig. 2 shows the inference path. A request carries the answers of one endpoint form. A schema layer validates every field against its unit and admissible range, maps categorical answers to the coding of the training cohort, and records a zero entered for a biomarker that cannot physiologically be zero (for example insulin or skin-fold thickness) as unrecorded, matching the convention of the training data. The resulting typed biomarker vector, with value and availability, is passed to the four ensemble members of the endpoint; their probabilities are averaged into the risk score, and the request, the model version and the score are written to a versioned inference record. For the coronary endpoint the value-of-information controller of Section VI ranks the tests that have not yet been performed.

B. Clinical input forms

Table I lists the fields collected for each endpoint. The deployed model of each endpoint is trained on exactly these fields over its complete reference cohort; the coronary model additionally uses the three auxiliary heart cohorts (Section V). The coronary form encodes categorical answers as one-hot groups, which the schema layer converts to the cohort coding.

C. Prediction service

The trained ensembles are served by an HTTP service with four routes: `GET /schema/{endpoint}` returns the field definitions of Table I with units, ranges and options; `POST /predict/{endpoint}` accepts a form and returns the risk score; `POST /predict/heart/onehot` accepts the one-hot coronary form; and `POST /acquire/heart` accepts the coronary fields recorded so far and returns the risk score together with the net value of information, Eq. (11), of every test not yet performed. A test counts as performed when all biomarkers it reveals are present, and the controller is the one evaluated under protocol P3, fitted on the full Cleveland cohort. Every prediction is appended to a JSON-lines inference record that stores the request, the score and the model version, defined as a prefix of the SHA-256 digest of the serialized artifact. Each ensemble is stored as a single serialized artifact. TabPFN is persisted through its in-context training table and re-conditioned when the artifact is loaded, HARBOR through its parameter arrays and fitted token encoders, and CatBoost and logistic regression through their fitted estimators. After a one-time load of 5–24 s per endpoint, a request is answered in 0.24–0.46 s on two CPU cores.

D. Software verification

The repository contains unit and integration tests (Section XI) for: the zero gradient of the ℓ_1 attention penalty and the descent behavior of the entropy penalty (Propositions 2 and 3); mask integrity, i.e. invariance of every HARBOR variant to the stored value of an unobserved biomarker; the proper-score inequality of Proposition 4; agreement of the form coding with the cohort coding for all five endpoints; partition and grouping properties of the resampling protocols; and end-to-end requests to the service, including monotonic response of each risk score to its dominant marker, completeness of the inference record, and the ranking and stopping behavior of the test-acquisition route.

VIII. Experimental Setup

A. Cohorts

We evaluate the five endpoints on their reference cohorts (Table II). Values recorded as zero where zero is physiologically

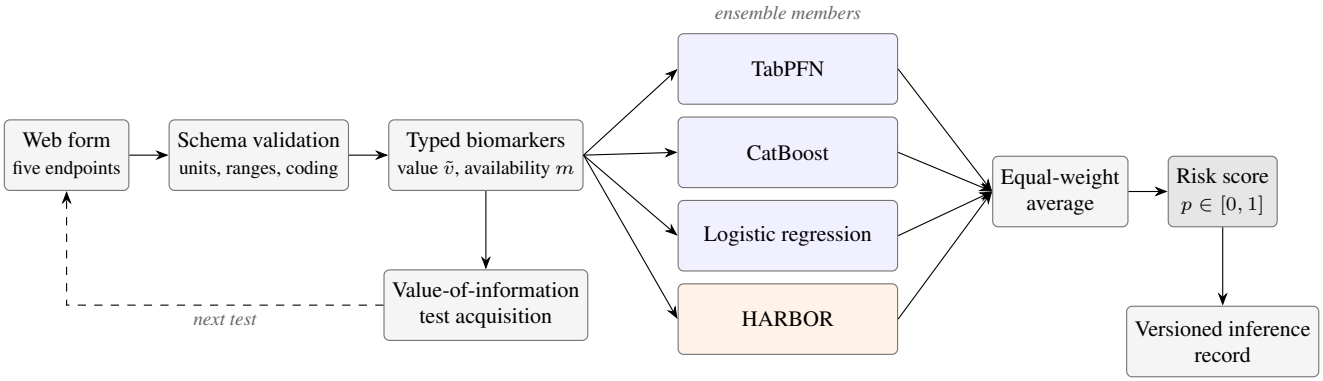


Fig. 2. PredictX inference path. Form answers are validated and coded into typed biomarkers (value and availability), scored by four heterogeneous learners whose probabilities are averaged into the risk score, and logged with the model version. The acquisition controller proposes the next test for the coronary endpoint.

TABLE I
FIELDS OF THE FIVE CLINICAL INPUT FORMS

Endpoint	Fields (unit)
Coronary disease	age (years), resting systolic blood pressure (mmHg), total cholesterol (mg/dL), maximum exercise heart rate (beats/min), exercise-induced ST depression (mm), sex, chest-pain type (4), fasting blood sugar >120 mg/dL, resting ECG (3), exercise-induced angina, ST-segment slope (3), vessels colored by fluoroscopy (0–3), thallium perfusion result (3)
Diabetes	pregnancies, plasma glucose (mg/dL), diastolic blood pressure (mmHg), triceps skin-fold (mm), 2-h serum insulin (μU/mL), body-mass index (kg/m^2), diabetes pedigree function, age (years)
Breast malignancy	clump thickness, uniformity of cell size, uniformity of cell shape, marginal adhesion, single epithelial cell size, bare nuclei, bland chromatin, normal nucleoli, mitoses (cytology scores 1–10)
Liver disease	total bilirubin (mg/dL), direct bilirubin (mg/dL), alkaline phosphatase (U/L), alanine aminotransferase (U/L), total protein (g/dL), albumin (g/dL), albumin/globulin ratio
Chronic kidney disease	blood pressure (mmHg), urine specific gravity, urine albumin grade (0–5), urine sugar grade (0–5), urine red blood cells, urine pus cells, pus-cell clumps

TABLE II
COHORTS

Endpoint (cohort)	n	Feat.	Pos.	Incompl.
Coronary disease (Cleveland [25])	303	13	46%	2%
Diabetes (Pima [26])	768	8	35%	49%
Breast malignancy (Wisconsin [27])	699	9	34%	2%
Liver disease (ILPD [28])	583	10	71%	1%
Chronic kidney disease (UCI CKD [29])	400	24	62%	60%
Aux. heart cohorts (HU, CH, VA)	617	13	60%	>99%

impossible (cholesterol and blood pressure in the heart cohorts; glucose, diastolic pressure, skin-fold, insulin and body-mass index in the Pima cohort) are recoded as unrecorded, following the cohort documentation, and every unrecorded value is passed to each learner in the form it supports natively. The benchmark uses all predictors of each cohort; the deployed configuration (Section E) uses the form fields of Table I, which coincide with the full predictor set for the coronary, diabetes and breast cohorts and comprise 7 of 10 liver and 7 of 24 kidney predictors.

B. Comparators

The *literature-standard methods* are the seven model families used in published comparisons on these cohorts: logistic regression, RBF support vector machine, k-nearest neighbors, Gaussian naive Bayes, random forest (500 trees), multilayer perceptron and XGBoost [6]. Non-tree methods use median imputation with missing-value indicators, standardization and one-hot encoding. Hyperparameters are selected by inner cross-validation (regularization strength for logistic regression; C and γ for the SVM; k for k-NN; weight decay for the MLP), and XGBoost is tuned with 25 Optuna [30] trials of inner 3-fold cross-validation over seven hyperparameters. The *state-of-the-art learners* are CatBoost with default settings [7] and TabPFN v2 with eight estimators [15]. HARBOR uses width $d = 32$, two layers, AdamW (learning rate 3×10^{-3}, weight decay 10^{-4}), full-batch training for at most 400 epochs with early stopping (patience 40) on a stratified 20% split, $\lambda_g = 10^{-3}$, and five ensemble members. On the public cohorts we set $\alpha_d = \kappa = \lambda_r = \lambda_c = 0$ and use the gate input $[h^{(0)}; m]$. HARBOR and the ensemble use the same hyperparameters on every cohort.

C. Protocols and statistics

P1, repeated cross-validation. Stratified 5-fold cross-validation repeated five times gives 25 test folds per cohort. Every fitted transformation is fitted inside the training folds, and the breast cohort is split by sample identifier so that repeated samples of one patient stay in the same fold. Paired AUROC differences are summarized with the corrected resampled t interval of Nadeau and Bengio [31], which accounts for the overlap of training sets across folds.

P2, published benchmark protocol. For a direct comparison with the strongest published results we reproduce the protocol of the TabPFN benchmark [14]: five stratified 50/50 train–test

TABLE III

DISCRIMINATION UNDER P1: MEAN AUROC OVER 25 TEST FOLDS. BOLD: HIGHEST MEAN (TIES WITHIN 0.0005). UPPER BLOCK: LITERATURE-STANDARD METHODS; LOWER BLOCK: STATE-OF-THE-ART LEARNERS, HARBOR AND PREDICTX.

Model	Coronary	Diabetes	Breast	Liver	Kidney
Logistic regression	0.909	0.834	0.994	0.748	**1.000**
SVM (RBF)	0.907	0.836	**0.995**	0.664	**1.000**
k-NN	0.901	0.814	0.992	0.720	0.994
Naive Bayes	0.865	0.801	0.980	0.729	0.990
Random forest	0.907	0.833	0.992	0.745	**1.000**
MLP	0.850	0.802	0.987	0.587	0.986
XGBoost (tuned)	0.905	0.840	0.992	0.722	**1.000**
CatBoost	0.903	0.831	0.992	0.746	**1.000**
TabPFN v2	**0.913**	**0.848**	**0.995**	**0.752**	**1.000**
HARBOR	0.909	0.840	0.992	0.742	**1.000**
PredictX	**0.913**	0.846	0.994	0.750	**1.000**
PredictX + cross-site	**0.915**	–	–	–	–

TABLE IV

MEAN PAIRED AUROC DIFFERENCE BETWEEN PREDICTX AND EACH COMPARATOR OVER 25 FOLDS (CORONARY: PREDICTX WITH CROSS-SITE TRAINING). $^{\uparrow}$: 95% NADEAU–BENGIO INTERVAL ENTIRELY ABOVE ZERO; $^{\downarrow}$: ENTIRELY BELOW ZERO. LAST COLUMN: WINS/TIES/LOSSES.

Comparator	Coronary	Diabetes	Breast	Liver	Kidney	W/T/L
Logistic regression	$+0.005$	$+0.012^{\uparrow}$	$+0.000$	$+0.002$	$+0.000$	1/4/0
SVM (RBF)	$+0.007$	$+0.011$	-0.001	$+0.087$	$+0.000$	0/5/0
k-NN	$+0.014$	$+0.032^{\uparrow}$	$+0.003$	$+0.030$	$+0.006^{\uparrow}$	2/3/0
Naive Bayes	$+0.050^{\uparrow}$	$+0.045^{\uparrow}$	$+0.014^{\uparrow}$	$+0.021$	$+0.010$	3/2/0
Random forest	$+0.007$	$+0.013^{\uparrow}$	$+0.003^{\uparrow}$	$+0.005$	$+0.000$	2/3/0
MLP	$+0.065$	$+0.045$	$+0.007$	$+0.164^{\uparrow}$	$+0.014^{\uparrow}$	2/3/0
XGBoost (tuned)	$+0.010$	$+0.007$	$+0.003$	$+0.028^{\uparrow}$	$+0.000$	1/4/0
CatBoost	$+0.012$	$+0.016^{\uparrow}$	$+0.002$	$+0.005$	$+0.000$	1/4/0
TabPFN v2	$+0.001$	-0.002	-0.001	-0.001	$+0.000$	0/5/0
HARBOR	$+0.005$	$+0.006$	$+0.002$	$+0.008$	$+0.000$	0/5/0

splits with ROC AUC averaged over splits. In that benchmark every baseline received 60 minutes of hyperparameter optimization per split.

P3, test acquisition. On the Cleveland cohort every patient starts from the baseline work-up (age, sex, chest-pain type, resting blood pressure, cholesterol, fasting glucose, resting ECG). Three tests can be ordered: exercise ECG (maximum heart rate, exercise angina, ST depression, ST slope; cost 1), fluoroscopy (number of opacified vessels; cost 2) and thallium scintigraphy (perfusion result; cost 2). Policies are evaluated on the P1 splits.

P4, controlled missingness. Forty percent of every numeric feature of two complete cohorts, Wisconsin Diagnostic Breast Cancer (WDBC, 30 features) [32] and ILPD (age and sex retained), is removed either completely at random (MCAR) or by self-masking (MNAR), with removal probability $\sigma(2z + b)$ in the feature's own z-score and b set to give a 40% rate. Single-member HARBOR variants and logistic regression are compared under P1.

IX. RESULTS

A. Benchmark

Table III reports mean AUROC under P1. PredictX attains the highest AUROC on coronary disease (0.915 with cross-site training) and ties at 1.000 on chronic kidney disease. On diabetes (0.846), breast malignancy (0.994) and liver disease (0.750) it is within 0.002 of the best learner, TabPFN v2. The strongest literature-standard method varies by cohort: logistic regression for coronary and liver disease, tuned XGBoost for diabetes, the SVM for breast malignancy and random forest for kidney disease.

Table IV tests every pairwise difference with corrected intervals. Across the 50 cohort–comparator pairs, PredictX is significantly better in 12 and significantly worse in none. It significantly outperforms naive Bayes on three cohorts; k-NN, random forest and the MLP on two; and logistic regression, tuned XGBoost and CatBoost on one each. Against the best literature-standard method of each cohort the mean differences are $+0.006$ for coronary disease (95% CI -0.010 to $+0.021$), $+0.007$ for diabetes (-0.003 to $+0.017$), $+0.002$ for liver disease (-0.019 to $+0.023$) and -0.001 for breast malignancy (-0.002 to $+0.001$) (Fig. 3).

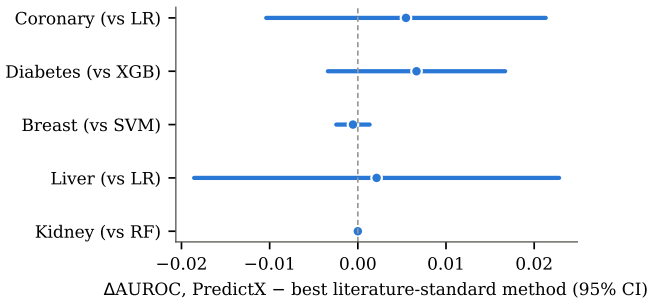


Fig. 3. PredictX minus the best literature-standard method of each cohort: mean paired AUROC difference with 95% Nadeau–Bengio interval over 25 folds.

TABLE V
ACCURACY (%, THRESHOLD 0.5) AND BRIER SCORE: BEST LITERATURE-STANDARD METHOD VERSUS PREDICTX (P1, 25 FOLDS)

		Best standard		PredictX	
Cohort	Method	Acc.	Brier	Acc.	Brier
Coronary	Logistic regression	84.1	0.119	83.8	0.117
Diabetes	XGBoost (tuned)	75.7	0.156	76.3	0.153
Breast	SVM (RBF)	96.4	0.026	96.7	0.025
Liver	Logistic regression	72.0	0.173	71.9	0.171
Kidney	Random forest	99.8	0.009	100.0	0.001

Table V compares accuracy at the 0.5 threshold and the Brier score with the best literature-standard method of each cohort. PredictX has the lower Brier score on all five cohorts (coronary 0.117 versus 0.119, diabetes 0.153 versus 0.156, breast 0.025 versus 0.026, liver 0.171 versus 0.173, kidney 0.001 versus 0.009), consistent with Proposition 4, and its accuracy ranges from 71.9% (liver) to 100% (kidney). Among all eleven models, PredictX has the lowest Brier score on liver disease and the second lowest, after TabPFN v2, on the other four cohorts.

B. Comparison with published results

Table VI places our results beside the TabPFN benchmark [14], in which every baseline received 60 minutes of hyperparameter optimization per split, under its 50/50 protocol (P2). PredictX exceeds the best published value on breast malignancy (0.9953 versus 0.9940) and liver disease (0.7438 versus 0.7412), and its TabPFN v2 member exceeds the best published diabetes value (0.8463 versus 0.8427). The PredictX ensemble scores 0.8402 on diabetes under this protocol.

C. Ablations

Ensemble members. Removing a single member changes mean AUROC by at most 0.0023 (Table VII). The largest contributions are TabPFN v2 on diabetes (+0.0023), logistic regression on coronary disease (+0.0018) and TabPFN v2 on coronary disease (+0.0014); HARBOR contributes +0.0009 on diabetes.

Cross-site training. Adding the Hungarian, Swiss and VA Long Beach cohorts to every training fold raised the mean coronary AUROC of logistic regression (+0.002), CatBoost (+0.001), HARBOR (+0.005) and PredictX (+0.001), and left TabPFN v2 at 0.913 (Table VIII). HARBOR, whose typed tokens represent the auxiliary cohorts' missing tests explicitly, gains most.

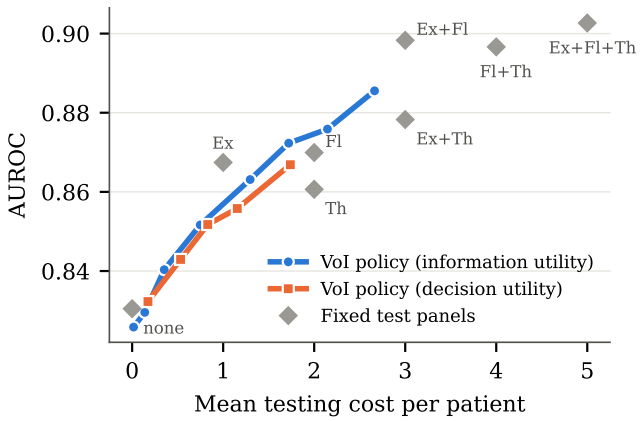


Fig. 4. Testing cost versus AUROC for coronary disease (P3, 25 folds). Grey diamonds: fixed test panels (Ex exercise ECG, Fl fluoroscopy, Th thallium scintigraphy). Lines: value-of-information policies over a sweep of λ.

Availability channel. Table IX reports protocol P4. Under MCAR the pattern is independent of the label by construction (mask-only AUROC 0.573 on WDBC and 0.433 on ILPD for the injected realization), and the three HARBOR variants lie within 0.005 of each other. Under self-masking MNAR on WDBC, where the mask alone predicts malignancy with AUROC 0.969, the variants that represent availability reach 0.979 (two-channel) and 0.981 (mask-concat), against 0.966 for the value-only encoder; the two-channel minus value-only difference is +0.013 (95% CI −0.001 to +0.027). On ILPD, where the injected pattern is weakly informative (mask-only AUROC 0.626), the variants lie within 0.002 of each other.

D. Value-of-information test acquisition

Fig. 4 and Table X report protocol P3. With $\lambda = 0$, the information-utility policy orders 1.53 of the three tests per patient at a mean cost of 2.66, against 5.0 for the full panel, and reaches AUROC 0.886 and accuracy 82.2%, against 0.903 and 83.2% with every test: 98% of the full-panel AUROC at 53% of the testing cost. Increasing λ traces a monotone frontier down to the baseline work-up (AUROC 0.830 at zero cost). Above a mean cost of 0.2 the information utility attains a higher AUROC than the decision utility at matched cost, because the decision utility values a test exactly when its result can change the current decision. Among fixed panels, exercise ECG with fluoroscopy reaches 0.898 at cost 3.0.

E. Deployed models on the clinical input forms

Table XI reports the deployed configuration: PredictX trained on exactly the form fields of Table I, evaluated under P1. The deployed models reach AUROC 0.915 for coronary disease, 0.846 for diabetes, 0.994 for breast malignancy, 0.741 for liver disease and 0.996 for chronic kidney disease, with accuracies between 70.2% and 97.8%. With seven routine urine and blood-pressure fields, the kidney model retains an AUROC of 0.996 relative to 1.000 with all 24 predictors.

TABLE VI
ROC AUC UNDER THE PUBLISHED 50/50 PROTOCOL (FIVE SPLITS). LEFT: VALUES PUBLISHED BY HOLLMANN ET AL. [14] (N.E.: WITHOUT ENSEMBLING). RIGHT: THIS WORK. BOLD: ROW MAXIMUM.

	Published [14]						This work		
Cohort	CatBoost	XGBoost	AutoGluon	TabPFN n.e.	TabPFN	TabPFN+AutoGluon	TabPFN v2	CatBoost	PredictX
Diabetes	0.8383	0.8378	0.8391	0.8412	0.8410	0.8427	**0.8463**	0.8217	0.8402
Breast	0.9931	0.9896	0.9933	0.9931	0.9934	0.9940	**0.9955**	0.9938	0.9953
Liver	0.7279	0.7171	0.7230	0.7412	0.7379	0.7326	0.7430	0.7326	**0.7438**

TABLE VII
LEAVE-ONE-MEMBER-OUT: AUROC OF PREDICTX MINUS PREDICTX WITHOUT THE LISTED MEMBER (MEAN OVER 25 FOLDS).

Cohort	TabPFN v2	CatBoost	LR	HARBOR
Coronary	+0.0014	−0.0012	+0.0018	+0.0000
Diabetes	+0.0023	−0.0011	+0.0009	+0.0009
Breast	+0.0004	−0.0003	+0.0003	−0.0002
Liver	+0.0000	+0.0011	+0.0001	−0.0015
Kidney	+0.0000	+0.0000	+0.0000	+0.0000

TABLE VIII
CROSS-SITE TRAINING ON CORONARY DISEASE: MEAN AUROC ON CLEVELAND TEST FOLDS WHEN TRAINING ON CLEVELAND ONLY AND ON CLEVELAND PLUS THE THREE AUXILIARY COHORTS.

Learner	Cleveland	+ auxiliary	Δ
Logistic regression	0.909	0.911	+0.002
CatBoost	0.903	0.904	+0.001
TabPFN v2	0.913	0.913	+0.000
HARBOR	0.909	0.914	+0.005
PredictX	0.913	0.915	+0.001

X. CONCLUSION

PredictX-HARBOR couples a typed, availability-aware biomarker representation with a two-channel graph encoder, a heterogeneous ensemble and a value-of-information acquisition controller, and serves the resulting models through validated clinical input forms. Across five disease cohorts the ensemble attains the highest coronary AUROC (0.915), is within 0.002 of the best learner on the remaining cohorts, is significantly better in 12 of 50 paired comparisons and significantly worse in none, and has a lower Brier score than the best literature-standard method on every cohort. Under the published benchmark protocol it sets the highest breast and liver results, and for coronary disease it reduces testing cost by 47% while retaining 98% of full-panel discrimination. The next step is a multi-hospital study whose assay, timing and quality metadata engage the quality-conditioned gate and the reviewed graph prior of Section IV.

XI. CODE AND DATA AVAILABILITY

All experiments, tables and figures are produced by the accompanying repository, whose layout is listed in Table XII. Library versions are pinned in `requirements.txt`. All five cohorts are publicly released, de-identified datasets (Table II), and no new patient data were collected; the TabPFN v2 weights are downloaded from the public Prior Labs release on first use and are used under the Prior Labs License 1.1 (`licenses/TabPFN-LICENSE.txt`). The deployed system is "Built with PriorLabs-TabPFN".

A complete reproduction runs `experiments.benchmark` for each cohort and protocol, `experiments.acquisition`, `experiments.missingness` and `experiments.report`; `python -m pytest` runs the test suite.

TABLE IX

CONTROLLED MISSINGNESS (P4; 40% OF EVERY NUMERIC FEATURE REMOVED). MASK ONLY: LOGISTIC REGRESSION ON THE MISSINGNESS PATTERN. HARBOR VARIANTS ARE SINGLE-MEMBER MODELS. LAST TWO COLUMNS: PAIRED AUROC DIFFERENCES WITH CORRECTED 95% INTERVALS OVER 25 FOLDS.

Data	Mech.	Mask only	LR	Value-only	Mask-concat	Two-channel	Two-ch. − Value-only	Two-ch. − Mask-concat
WDBC	MCAR	0.573	0.986	0.983	0.985	0.982	−0.001 [−0.005, +0.004]	−0.003 [−0.011, +0.005]
WDBC	MNAR	0.969	0.988	0.966	0.981	0.979	+0.013 [−0.001, +0.027]	−0.002 [−0.015, +0.011]
ILPD	MCAR	0.433	0.723	0.709	0.714	0.713	+0.004 [−0.014, +0.023]	+0.000 [−0.015, +0.014]
ILPD	MNAR	0.626	0.705	0.696	0.697	0.695	−0.001 [−0.026, +0.024]	−0.002 [−0.014, +0.011]

TABLE X

TEST-ACQUISITION POLICIES FOR CORONARY DISEASE (P3): MEAN COST, TESTS PER PATIENT, AUROC AND ACCURACY (%).

Policy	Cost	Tests	AUROC	Acc.
Fixed: none	0.00	0.00	0.830	76.8
Fixed: Ex	1.00	1.00	0.867	78.7
Fixed: Fl	2.00	1.00	0.870	77.8
Fixed: Th	2.00	1.00	0.861	77.4
Fixed: Ex+Fl	3.00	2.00	0.898	82.9
Fixed: Ex+Th	3.00	2.00	0.878	79.5
Fixed: Fl+Th	4.00	2.00	0.897	82.2
Fixed: Ex+Fl+Th	5.00	3.00	0.903	83.2
VoI (information), $\lambda = 0$	2.66	1.53	0.886	82.2
VoI (information), $\lambda = 0.02$	1.72	1.04	0.872	80.5
VoI (information), $\lambda = 0.05$	0.75	0.53	0.852	79.0
VoI (information), $\lambda = 0.12$	0.14	0.13	0.830	76.9
VoI (decision), $\lambda = 0$	1.74	1.01	0.867	80.4
VoI (decision), $\lambda = 0.02$	0.83	0.53	0.852	80.1

TABLE XI

DEPLOYED CONFIGURATION: PREDICTX ON THE FORM FIELDS (P1, 25 FOLDS), WITH THE BEST LITERATURE-STANDARD METHOD ON THE SAME FIELDS.

		PredictX		Best standard	
Endpoint	Fields	AUROC	Acc.	Method	AUROC
Coronary	13	0.915	83.8	Logistic regression	0.909
Diabetes	8	0.846	76.3	XGBoost (tuned)	0.840
Breast	9	0.994	96.7	SVM (RBF)	0.995
Liver	7	0.741	70.2	Logistic regression	0.745
Kidney	7	0.996	97.8	XGBoost (tuned)	0.996

TABLE XII

REPOSITORY LAYOUT

Path	Contents
`predictx/cohorts.py`	cohort loaders; full and form feature sets
`predictx/preprocessing.py`	fold-safe encoders; HARBOR token encoder
`predictx/harbor.py`	HARBOR encoder, variants, deep ensemble
`predictx/learners.py`	comparators, CatBoost, XGBoost, TabPFN v2
`predictx/ensemble.py`	PredictX ensemble
`predictx/acquisition.py`	value-of-information controller
`predictx/evaluation.py`	protocols, corrected resampled t-test
`predictx/website.py`	form schemas, conversion to model input
`experiments/benchmark.py`	predictions under P1 and P2
`experiments/acquisition.py`	protocol P3
`experiments/missingness.py`	protocol P4
`experiments/report.py`	all tables, figures and quoted numbers
`service/train_models.py`	deployed ensembles on the form fields
`service/app.py`	HTTP prediction service
`tests/`	unit and end-to-end tests
`licenses/`	third-party model license
`results/`	per-split predictions of every model

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